INDIVIDUAL PROJECT REPORT

ON

“Arya College ClubSphere: Centralized Club and Event Management Platform”

INDIVIDUAL CONTRIBUTION REPORT — EVENT MANAGEMENT & QR CODE REGISTRATION

Submitted in partial fulfillment of the requirement for the award of Degree of

Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

Submitted by

AKANSHA SHARMA

Enrollment No.: 24EARCS012

Individual responsibility: Event Management & QR Code Registration

PBL Team ID: PBL2627-CSE-A-024

Under the guidance of

Er. Ram Babu Buri

Department of Computer Science & Engineering

Session: 2026–27

ARYA COLLEGE OF ENGINEERING & I.T.

SP-42, RIICO Industrial Area, Kukas, Jaipur

CERTIFICATE

This is to certify that the individual project report entitled “Arya College ClubSphere — Event Management & QR Code Registration” has been prepared by Akansha Sharma (24EARCS012), a student of Bachelor of Technology in Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, affiliated with Rajasthan Technical University, Kota, during the academic session 2026–27.

The report records the student’s individual contribution to the group project Arya College ClubSphere (PBL2627-CSE-A-024), carried out under the guidance of Er. Ram Babu Buri. The contribution described here forms part of a four-member team project; the work of the other team members is acknowledged and is not claimed in this report.

Role	Name	Signature
Project Guide	Er. Ram Babu Buri Dept. of CSE, ACEIT, Jaipur	
Head of Department	Dr. Akhil Pandey Dept. of CSE, ACEIT, Jaipur	
Student	Akansha Sharma (24EARCS012)	

Date: _____ Place: Jaipur

DECLARATION

I, Akansha Sharma (24EARCS012), a student of B.Tech. Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, hereby declare that this individual report describes my own contribution to the group project Arya College ClubSphere, carried out during the session 2026–27 under the guidance of Er. Ram Babu Buri.

The technical content of this report was prepared from the project source code, database schema and module documentation. Features built by teammates are stated as such. The report does not claim sole authorship of ClubSphere or measured production performance.

To the best of my knowledge, this report has not been submitted to any other university or institution for the award of any degree or diploma.

Akansha Sharma

Enrollment No.: 24EARCS012

B.Tech. (CSE), Session 2026–27

Signature: _____ Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Er. Ram Babu Buri, for his guidance throughout the planning and development of ClubSphere. His emphasis on clear requirements and honest reporting of status shaped the way I designed and documented the Event Management & QR Code Registration module.

I am thankful to Dr. Akhil Pandey, Head of the Department of Computer Science & Engineering, Arya College of Engineering & I.T., for the academic environment, and to the faculty of the department for their support.

I am grateful to my teammates — Avantika Mishra, Avanshi Kothari and Aditi Maheshwari. My module depends on a shared MySQL schema, AuthenticationFilter and JSP chrome. Working with Git taught me integration and communication within a team.

Finally, I thank my family and friends for their encouragement and patience during the project.

Akansha Sharma

Enrollment No.: 24EARCS012

ABSTRACT

This report documents my contribution to ClubSphere: Event Management and QR Code Registration. I implemented event create/edit/cancel for club heads, public event listing, individual and group registration, and QR PNG generation with ZXing. Each event has a unique qr_token. Scanning the code opens /event/register?token=… after login if needed. QRCodeUtilTest verifies PNG output.

Keywords: Event registration, QR code, ZXing, capacity, JDBC

TABLE OF CONTENTS

Certificate
Declaration
Acknowledgement
Abstract
CHAPTER 1 — INTRODUCTION
CHAPTER 2 — PROJECT OVERVIEW AND INDIVIDUAL RESPONSIBILITIES
CHAPTER 3 — TECHNOLOGY STACK
CHAPTER 4 — REQUIREMENTS OF MY MODULE
CHAPTER 5 — DESIGN AND IMPLEMENTATION
CHAPTER 6 — TESTING AND VERIFICATION
CHAPTER 7 — CHALLENGES AND LEARNING OUTCOMES
CHAPTER 8 — CONCLUSION AND FUTURE SCOPE
REFERENCES

CHAPTER 1 — INTRODUCTION

1.1 Background

ClubSphere is the group PBL project (PBL2627-CSE-A-024) that centralises clubs, events, QR registration, tasks and notifications for Arya College of Engineering & I.T. The team report describes the whole system. This individual report documents only my contribution: **Event Management & QR Code Registration**.

1.2 Purpose of this Report

This report separates my work from that of my teammates, explains the design of my module, how it was tested, and what remains incomplete.

1.3 Problem Addressed by My Module

Club heads needed a structured event record, and students needed a reliable signup path at a venue. A QR bound to a database token is simpler than circulating long URLs.

1.4 Objectives of My Contribution

- Create events with schedule, venue, capacity and participation type.
- Issue a unique qr_token per event.
- Render /qr/{token} as a PNG of the registration URL.
- Register once per user per event; store group participants.

CHAPTER 2 — PROJECT OVERVIEW AND INDIVIDUAL RESPONSIBILITIES

2.1 Overall Project Context

ClubSphere is developed by four students and hosted at github.com/Avantikamishra862/Arya_Clubsphere. It is a Java 17 / JSP / Servlet / MySQL application deployed as a Tomcat WAR.

2.2 Module Ownership

Member	Enrollment	Module
Avantika Mishra	24EARCS032	Authentication & User Management
Avanshi Kothari	24EARCS031	Club Management
Akansha Sharma	24EARCS012	Event Management & QR Code Registration
Aditi Maheshwari	24EARCS007	Dashboards, Tasks & Notifications

My owned area is highlighted in the implementation chapter. Shared files (schema.sql, AuthenticationFilter, navbar.jsp, pom.xml) were coordinated with the team.

2.3 In-scope and Out-of-scope

In scope: EventServlet, RegistrationServlet, QRCodeServlet, EventService, RegistrationService, QRCodeUtil, event JSPs, club-head event form and QR view.

Out of scope: User login implementation and club roster CRUD belong to teammates; I consume currentUser and club_id.

CHAPTER 3 — TECHNOLOGY STACK

I used the shared ClubSphere stack. My module specifically depends on the items below.

Layer	Technology	How I used it
QR	ZXing 3.5.3	PNG 320×320, error correction H
Web	EventServlet, RegistrationServlet, QRCodeServlet	Catalogue, signup, image
Data	events, registrations, event_participants	Token, unique user+event
Test	QRCodeUtilTest	PNG magic bytes and data URI

CHAPTER 4 — REQUIREMENTS OF MY MODULE

4.1 Users

Club heads create events. Students and members register. Admins can list all events and registrations.

4.2 Functional Requirements

ID	Requirement
FR-E1	Create/edit event with qr_token
FR-E2	Public list/view with filters
FR-E3	Register by eventId or token
FR-E4	Reject duplicate registration
FR-E5	Group participant rows
FR-E6	Downloadable QR PNG

4.3 Non-functional requirements

APP_BASE_URL must match the public origin (WAR context /aryaClgClubSphere versus Docker ROOT /). QR generation should fail closed with HTTP 404 if the token is unknown.

CHAPTER 5 — DESIGN AND IMPLEMENTATION

5.1 Request flow

ClubHeadServlet saves event → token stored → QRCodeServlet builds APP_BASE_URL/event/register?token= → student RegistrationServlet → RegistrationService insert and registered_count++.

5.2 Principal classes and views

Layer	Items
Controllers	EventServlet, RegistrationServlet, QRCodeServlet, ClubHeadServlet event paths
Service	EventService, RegistrationService, QRCodeUtil
Views	events/list.jsp, view.jsp, register.jsp, confirmation.jsp, club-head/event-qr.jsp

5.3 Implementation notes

Lookup accepts token or eventId. Unauthenticated users are sent to login with a return URL. Group forms store event_participants. Cancelled events are a club-head action.

UML figures for the whole system are in the team report and in docs/UML/Arya_College_ClubSphere_UML_Diagrams.pdf.

CHAPTER 6 — TESTING AND VERIFICATION

6.1 Strategy

QRCodeUtilTest plus manual create-event, download QR, register as student_1, duplicate reject, admin registration list.

6.2 Representative checks

Check	Result / note
PNG magic bytes	Unit test passed
Duplicate register	Confirmation of existing record
Unknown token	404 / not found message
QR download=true	Content-Disposition attachment

CHAPTER 7 — CHALLENGES AND LEARNING OUTCOMES

7.1 Challenges

Absolute URLs inside QR codes versus different deploy context paths; individual and group forms on one servlet; keeping capacity and deadline fields consistent across JSPs.

7.2 Learning outcomes

I learned to treat QR content as a configured public URL, not a hard-coded localhost path, and to enforce uniqueness both in SQL and in the service.

CHAPTER 8 — CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Events and QR registration are the student-facing core of ClubSphere. Binding tokens to events keeps check-in links simple: scan, log in if needed, confirm.

8.2 Future work in my module

Attendance scan distinct from registration QR, waitlists when capacity is full, and email tickets.

REFERENCES

[1] Jakarta Servlet 6.0 Specification.

[2] Apache Tomcat 10.1 Documentation.

[3] MySQL 8.0 Reference Manual.

[4] BCrypt password hashing documentation.

[5] ZXing QR code library documentation.

[6] Arya College ClubSphere team project report, Dept. of CSE, ACEIT, Jaipur, Session 2026–27, PBL2627-CSE-A-024.

[7] https://github.com/Avantikamishra862/Arya_Clubsphere